Student's Full Name: ……..

THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM ONE SEPTEMBER MID – TERM EXAMINATIONS

033 **BIOLOGY**

(For Both Private and School Candidates)

Time: 2:30 Hours **September: 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **ten (10)** questions.
2. Answer **all** questions in all sections.
3. Sections A and C carry **fifteen (15)** marks each, where section B carries **seventy (70)** marks.
4. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
5. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
6. Write your **Name in Full** on every page of your answer sheet(s).

QUESTION NUMBER	FOR EXAMINER'S USE ONLY	
	MARKS	EXAMINER'S INITIALS
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		

BIOS F1 MID TERM EXAM 2025

SECTION A (15 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer spaces provided.

(i) Which of the following sets of things would biologists prefer to study in details?

A Rat, car, tortoise and human being
B Louse, baobab tree, ruler and bacteria
C Ruler, computer, car and television
D Flamingo, penguin, cockroach and wildebeest

(ii) Rosemary Denis observed a new cell. The cell had no true nucleus. It only had the genetic material that is not protected by a nuclear membrane. The cell is most likely to be:

A The animal cell
B The eukaryotic cell
C The plant cell
D The bacterial cell

(iii) Which among the following is the list of animal tissues?

A Muscle, blood and bone
B Skin, heart and brain
C Brain, bone and liver
D Muscle, liver and skin

(iv) The microscope has an eyepiece with a magnifying power of 10x and the objective lens of 40x. What is the total magnification of this microscope?

A 30x
B 50x
C 4x
D 400x

(v) Which of the following represent presentations of complex biological systems into simple pictures or diagrams?

A Models
B Charts
C Cages
D Aquaria

(vi) When a bread is left for three weeks in a wet place, the certain greyish structures grow on it. These structures are known as:

A Mushrooms
B Moss plants
C Yeast
D Bread molds

(vii) Aniceta wants to know how to maintain a good health. From your knowledge of biology, what can you advise her?

A Sleeping without treated mosquito net
B Eating balanced diet like chips, chocolate and yoghurt
C Washing hands with water and soap before eating food and after visiting the toilet
D Doing heavy exercises every day

(viii) Tics live on the skin of cows and suck blood from them. Therefore, tics are:

A Parasites
B Non-living things
C Microorganisms
D Insects

(ix) Which of the following is correct about the leaf of hibiscus plant?

A It is short and broad B It is thin and long

C It has parallel veins D It is yellow in colour

(x) Nancy Aldin always complains of a burning pain in the stomach and gives out blood-stained poop. The doctor confirmed that she is suffering from stomach ulcers. Which of the following pathogens is Nancy affected off?

A *Helicobacter pylori* B *Clostridium tetani*

C *Streptococcus pneumoniae* D *Campylobacter jejuni*

ANSWERS

S/N	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
ANS										

2. Match the description in List A with the respective items in List B by writing the letter of the correct response besides the item number in the answer spaces provided.

LIST A	**LIST B**
(i) A vessel in plants that transports manufactured food from leaves to other parts.	A Autotrophs
(ii) A hole or opening found between two adjacent guard cells on the leaves in green plants used for gaseous exchange.	B Heterotrophs
(iii) Organisms which cannot manufacture their own food; for example, human beings.	C Cytoplasm
(iv) The form in which plants store food that they manufacture.	D Phloem
(v) A vessel in plants that transports water and dissolved salts from roots to other parts.	E Xylem
	F Stoma
	G Starch
	H Root hair cell

ANSWERS

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B					

SECTION B (70 Marks)

Answer **all** questions in this section

3. (a) What do you understand by the following terminologies?

(i) Tissue ……

……

(ii) Laboratory ……

……

(iii) First aid ……

……

(iv) Zoology ……

……

(b) Biology is revealed in every aspect of life. Explain four activities that we normally do or see which involve biological concepts.

(i) …….
(ii) …….
(iii) …….
(iv) …….

4. (a) Identify the field and branch of biology that you will use to solve the following problems:

(i) Protecting cattle like cows from flies like tsetse flies.

Field: …….
Branch: …….

(ii) Conserving village river from drying.

Field: …….
Branch: …….

(iii) Helping a person with a broken leg.

Field: …….
Branch: …….

(b) Outline the importance of planting trees in our environment by giving out four points.

(i) …….
(ii) …….
(iii) …….
(iv) …….

5. (a) Distinguish between hypothesis and experiment.

……..
……..
……..
……..

(b) Explain three types of variables used during biological experiments.

(i) …….
……
(ii) …….
……
(iii) …….
……

(c) What is the main aim of conducting experiment?

……..

(d) Complete the table below about common measurements involved in biological studies:

MEASUREMENT	INSTRUMENT	SI – UNIT
Length		
Temperature		

6. (a) Why do we need to have the laboratory at the school?

……..
……..

(b) Give the reason for each of the following facts:

(i) We should not eat or drink in the laboratory.

……
……

(ii) We should open all windows when we are in the laboratory.

……
……

(iii) The laboratory should have the first aid kit.

……
……

(iv) Teachers should teach biology in the laboratory instead of a normal classroom.

……
……

(v) It is difficult to observe cells if we don't have the microscope.

……
……

(c) Give the names of the following:

(i) Apparatus used to catch small aquatic animals ……..
(ii) Apparatus used to scoop powdery or crystalline substances ……..
(iii) Any change in the environment that is detected by an organism. ……..
(iv) A person who is specialized in the study of living things. ……

7. (a) Briefly explain why:

(i) Plants do not need mouth to eat food like human beings. ……..
……

(ii) Plant leaves are usually green in colour. ……
……

(b) What is the difference between nutrition and photosynthesis?

……..
……..
……..
……..

(c) What are the raw materials needed for photosynthesis to take place?

(i) ……
(ii) ……

(d) Explain three importance of photosynthesis for the continuous survival of plants and other organisms on earth.

(i) ……
(ii) ……
(iii) ……

8. (a) What is cell differentiation? ..……..
……..
……..
……..

(b) Give the reason for why:

(i) Mature red blood cell does not have nucleus. ...…….
……
……

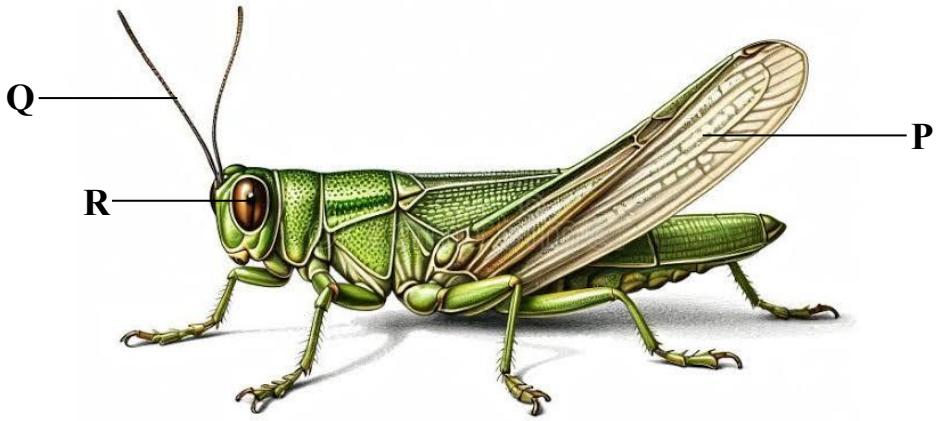
(ii) Sperm cell has a tail. ...…….
……
……

(iii) Animal cell bursts if it stays in pure water for long time, but plant cell does not burst.
……
……
……

(c) (i) Give the name of the cell that is specialized for transmission of information in in the body in form of electrical signals.
……

(ii) Draw a neat diagram of the cell named in (i) above and label its four parts.

9. Form one students at Kindoroko girls' secondary school collected different specimens in order to study them. One of the specimens collected is shown below. Study it carefully and then answer the questions that follow.



Specimen AA

(a) (i) What is the common name of specimen **AA** above? ……

(ii) Classify specimen **AA** from Kingdom to class level.

Kingdom	Phylum	Class

(b) (i) In which places do we find specimen **AA**? ...……

(ii) What are the functions of parts P, Q and R?

P ……

Q ……

R ……

(c) What is the main advantage of specimen **AA**? ……..……..

……..

(d) When the head of specimen **AA** was dipped in water for 30 minutes, it did not die; but when its abdomen was dipped in water, it died just within 10 minutes. Why? ……

……..

SECTION C (15 Marks)

Answer the question number **ten (10)** below

10. Bacteria cannot be seen by naked eyes but yet they are considered as living things. How can you prove that they are living things? Give six points.

…….

…….

…….

…….

…….

…….

Kwa uhitaji wa msaada wowote katika mambo ya kitaaluma na somo la biology (O-Level; Theory & Practical) kama vile:

(i) Maswali yenye utata
(ii) Kuandaliwa marking scheme ya mtihani wako
(iii) Kuandaliwa mtihani na marking scheme yake
(iv) Kuandaliwa complete set ya mitihani (Masomo yote) mf. Pre National
(v) Kufanya joint za biology na wanafunzi wa shule ninayofundisha
(vi) Kuedit na moderate mtihani
(vii) Kuandaa mifumo ya Microsoft Excel ya kuchakata matokeo au rekodi na mahesabu ya biashara

Wasiliana na mimi **Mwalimu Ephraim Joseph** wa ***Shule ya Sekondari ya Wasichana Kindoroko*** iliyopo wilaya ya MWANGA mkoani KILIMANJARO inayomilikiwa na Jimbo Katoliki la Same.

- Simu (Call/SMS/Whatsapp): - **0621660890**
- TikTok: @ephraim_seth (Tunaokolewa kwa Neema)

THE CATHOLIC DIOCESE OF SAME
KINDOROKO GIRLS' SECONDARY SCHOOL
FORM ONE SEPTEMBER MID – TERM EXAMINATIONS [SEPT 2025]

033 **BIOLOGY**

(For Both Private and School Candidates)

MARKING GUIDE

1. 10 Marks @1

QUESTION	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
ANSWER	**D**	**D**	**A**	**D**	**B**	**D**	**C**	**A**	**A**	**A**

2. 5 Marks @1

LIST A	(i)	(ii)	(iii)	(iv)	(v)
LIST B	**D**	**F**	**B**	**G**	**E**

3. (a) (i) A group of several cells that work together for a specific function
   (ii) A special room or building designed for conducting scientific experiments.
   (iii) Immediate help given to a sick or injured person before seeking for advanced medical help.
   (iv) The branch of biology that deals with the study of animals. **(6 Marks @1.5)**
   (b) (i) Crop cultivation
   (ii) Cleaning of the environment
   (iii) Cooking and eating food
   (iv) Caring for children, elders and sick people **(4 Marks @1)**

4. (a) (i) *Field:* Agriculture, Veterinary science
   Branch: Parasitology, Ecology, Zoology
   (ii) *Field:* Forestry
   Branch: Ecology
   (iii) *Field:* Medicine
   Branch: Anatomy, Physiology
   (b) (i) They retain water in the soil
   (ii) They prevent soil erosion
   (iii) They provide oxygen and reduce carbon dioxide
   (iv) They provide food such as fruits **(10 Marks @1)**

5. (a) **Hypothesis**: Is the suggested (possible) answer to the question asked. **WHILE**
   Experiment: Is a scientific procedure carried out under controlled conditions to determine whether the hypothesis is correct or not.

(b) (i) Dependent variable: Condition or factor that is being measured or tested
(ii) Independent variable: Condition or factor that is changed systematically to obtain different results.
(iii) Controlled variable: Condition or factor that is kept constant throughout the experiment.

(c) To determine whether the hypothesis is correct or not. (To test or confirm the hypothesis)

(d) *Common measurements:*

MEASUREMENT	INSTRUMENT	SI – UNIT
Length	Tape measure, Meter rule, Ruler	Meter (m)
Temperature	Thermometer	Kelvin (K)

(10 Marks @1)

6. (a) So as to conduct experiments and study science subjects practically.

(b) (i) So as to avoid poisoning.
(ii) So as to get good ventilation (to avoid suffocation)
(iii) So as to provide first aid for the victims in case of accidents. (So as to learn practically how to provide first aid)
(iv) So as to be able to perform the experiments of the concepts taught easily since specimens, chemicals and apparatii are present in the laboratory and not in the normal classroom.
(v) Because they are very small. Microscope is needed for magnification.

(c) (i) Fish net
(ii) Spatula
(iii) Stimulus
(iv) Biologist **(10 Marks @1)**

7. (a) (i) Because they make their own food through photosynthesis. **(3 Marks @1.5)**
(ii) Because they contain green pigments called chlorophyll for trapping sunlight.

(b) **Nutrition**: Is the process by which an organism obtains food from its environment necessary for survival. **WHILE**
Photosynthesis: Is the process by which plants make their own food by using carbon dioxide and water in presence of sunlight and chlorophyll. **(2 Marks @1)**

(c) (i) Carbon dioxide
(ii) Water **(2 Marks @1)**

(d) (i) Production of food
(ii) Production of fuel
(iii) Production of oxygen **(3 Marks @1)**

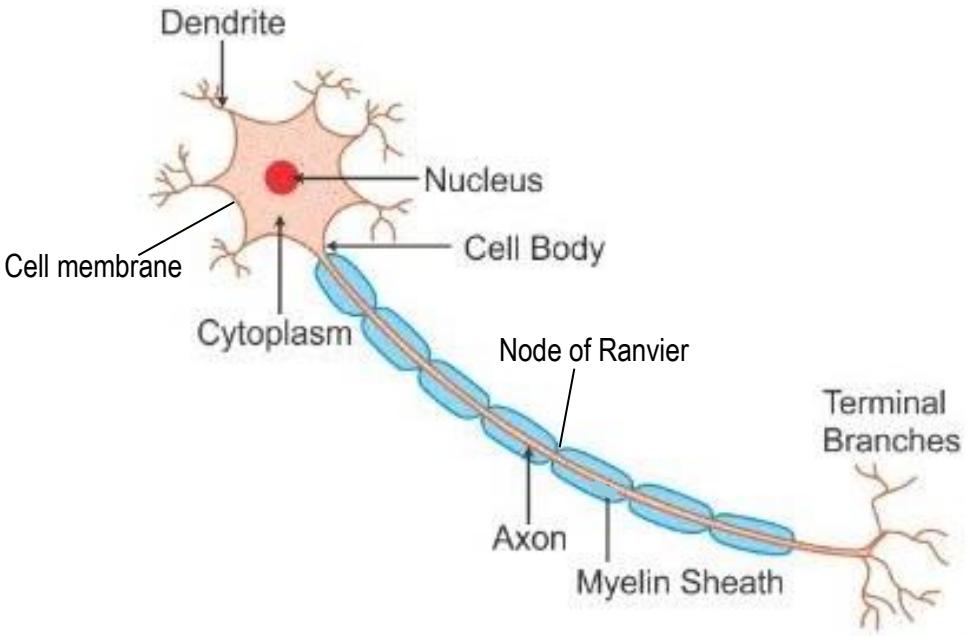
8. (a) Is the process by which cells undergo different changes in structure so as to be able to perform certain function. **(1 Mark)**

(b) (i) So as to have enough space for loading respiratory gases.
(ii) So as to be able to swim towards the egg
(iii) Because animal cells do not have cell walls like plant cell that prevents bursting after the cell absorbs much water (by osmosis) **(3 Marks @1)**

(c) (i) Nerve cell (Neuron) **(1 Mark)**

(ii) Caption = 1, Diagram = 2, Labels = 2 @0.5. ***(Tot: 5 Marks)***

THE DIAGRAM OF A NERVE CELL (NEURON)



9. (a) (i) Grasshopper

(ii) *Classification of specimen* **AA**.

Kingdom	Phylum	Class
Animalia	Arthropoda	Insecta

(b) (i) In areas with fresh grasses (Grass land)

(ii) **P** For flying (Locomotion)

Q For sensitivity

R For vision (Seeing)

(c) Used as food for other organisms like snakes, frogs, birds, humans **(10 Marks @1)**

(d) Because it breathes through spiracles which are found on its abdomen and not on the head

GUIDE FOR ESSAY QUESTION

Introduction:	2 Marks
Main body:	12 Marks (6 points @ 2)
Conclusion:	1 Mark

10. ***Introduction***: **<u>Meaning of living things or Bacteria</u>**

Living things: Are things that possess life. They are also called organisms. Example human being, trees, mushroom, insects, bacteria, etc.

OR

Bacteria: Are unicellular organisms with cell walls but do not have true nucleus. They are found in Kingdom monera. They are prokaryotics. Most of them are pathogenic (Cause diseases) but a few of them are non-pathogenic (They do not cause diseases)

Main body: **<u>Any six characteristics of living things</u>**.

The following are characteristics which prove that bacteria are living things.

(i) They feed.

Feeding (nutrition) is the process of obtaining food materials from the environment and absorb nutrients contained in it. Bacteria feed by decomposing dead organic matter and absorbing the simple nutrients

(ii) They respire.

This is the process of breaking down food materials within the cells to release energy. Bacteria respire since they contain mitochondria and can absorb glucose which is a raw material for respiration.

(iii) They excrete

This is the process of removing metabolic waste products from the body of living organisms. Bacteria remove most of their wastes by simple diffusion or by means of contractile vacuole.

(iv) They reproduce

This is the process by which living things produce new individuals of their own kind. Bacteria reproduce by mitosis where the parent bacterial cell divides to produce two identical offspring (Binary fission)

(v) They grow and develop

Growth is the permanent increase in size and dry mass of an organism. It is caused by the increase in number of cells that make up the organism. **Development** is increase in the complexity of an organism. It is caused by the change of body organs into more complex shape and form making an organism more complex. Bacteria cell are able to increase in size and ability to function.

(vi) They move

This is the act of changing position or posture. It can involve the whole body of an organism or some parts of the body. Bacteria are able to move by means of flagella or fimbriae (pilli)

(vii) They sense

This is the ability of an organism to detect and respond to the changes in the environment. Bacteria can detect different changes and respond to them accordingly

Conclusion: Importance of studying living things. [Or any other relevant conclusion]